

Class Test

- (a) Define Data Mining and KDD process

Data Mining: Data mining is the process of discovering useful, valid and previously unknown patterns or knowledge from large amounts of data.

Two Data Mining Tasks:

1. Classification
2. Clustering.

Steps in KDD

1. Data selection
2. Data cleaning
3. Data Integration
4. Data Transformation
5. Data Mining
6. Pattern Evaluation
7. Knowledge presentation.

Role of Data Preprocessing

1. Removes missing values and noise
2. Improves data quality and increases mining accuracy
3. Reduces processing time.

- (b) Data cleaning: Is the process of detecting and correcting incomplete, inaccurate, or inconsistent data.

Two Data Quality Issues:

1. Missing values
2. Noisy data

Handling Missing Values:

1. Ignore the record
2. Fill with mean/median/mode
3. Predict using machine learning.

Handling Noisy Data

Methods:

1. Binning
2. Regression
3. Clustering.

Q(a)

Data Integration: Data Integration is the process of combining data from multiple sources into a unified dataset.

Two challenges:

1. Data redundancy
2. Data inconsistency.

Handling Redundancy

1. Remove duplicate records
2. Perform correlation analysis.

Handling Inconsistency

1. standardize formats
2. Resolve conflicting values using business rules.

Q(b)

Covariance Analysis: covariance analysis measures how two variables change together.

Population covariance:

$$\text{cov}(x, y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

sample covariance

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

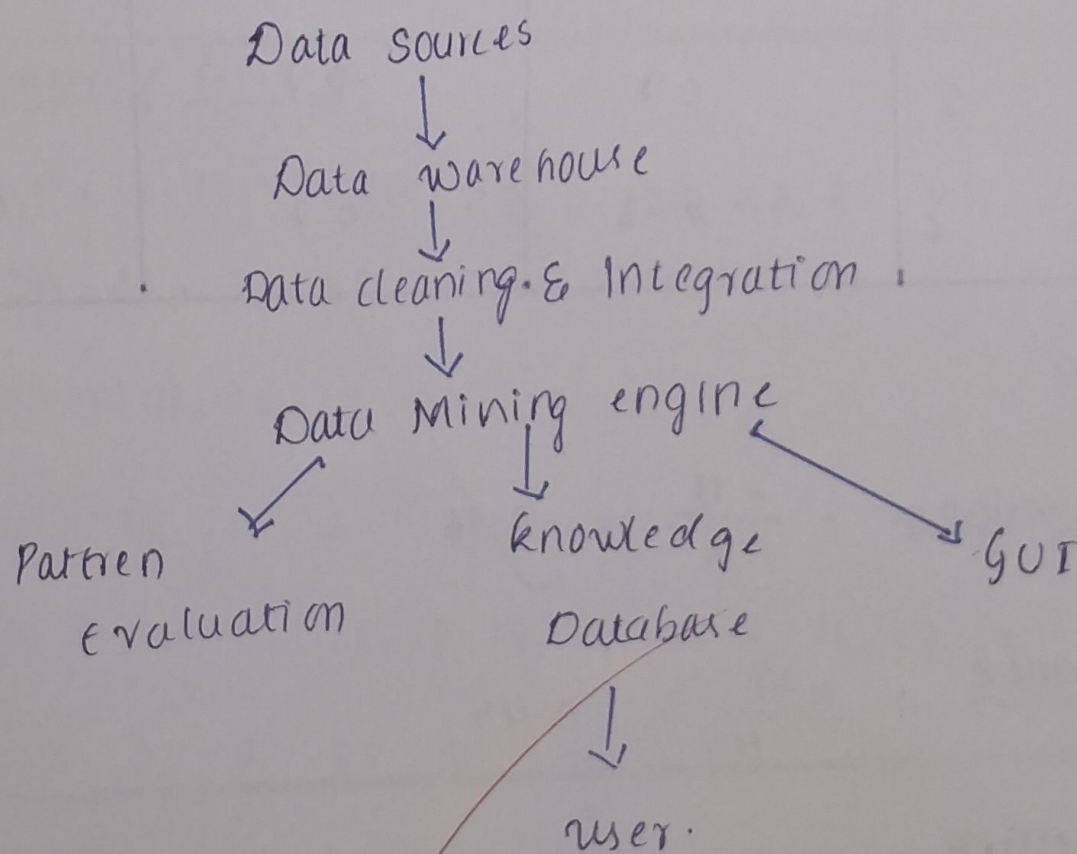
Stock Covariance

A	B
2	5
3	8
5	10
4	11
6	14

∴ Both stock values generally increase together

1. covariance is positive
2. The stocks rise together
3. They are not independent because they follow the same industry trend

(b) Architecture of Data Mining System



Interaction

1. Data is collected
2. cleaned and integrated
3. Mining engine discovers patterns
4. Pattern evaluation selects useful knowledge
5. GUI displays results to users

4(a)

Covariance calculation

$$x = \{4, 5, 6, 7, 9\}$$

$$y = \{8, 3, 7, 5, 6\}$$

$$\begin{aligned} \text{Mean of } x &= \frac{4+5+6+7+9}{5} \\ &= 6.2 \end{aligned}$$

$$\begin{aligned} \text{Mean of } y &= \frac{8+3+7+5+6}{5} \\ &= 5.8 \end{aligned}$$

x	y	(x - 6.2)	(y - 5.8)	product
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56

$$\text{sum} = -1.80$$

$$\text{Population covariance} : \frac{-1.8}{5} = -0.36$$

$$\text{Sample covariance} : \frac{-1.8}{4} = -0.45$$

4(b)

Data Preprocessing:Major steps:

1. Data cleaning

2. Data Integration

3. Data transformation

4. Data Reduction

5. Data Discretization

Preprocessing Pipeline

1. Remove missing values
2. Detect and remove outliers
3. Standardize data formats
4. Normalize data
5. Discretize if required.

Justification

Cleaning first improves quality, then consistency is ensured, followed by normalization and discretization for better mining performance

5(a)

Equal-frequency binning:

Data: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

$$\text{mean} = 18$$

Smoothed values: 18, 18, 18, 18, 18, 18, 18, 18, 18

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

$$\text{mean} = 28$$

Smoothed values: 28, 28, 28, 28, 28, 28, 28, 28, 28

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

$$\text{mean} = 44.9 \approx 45$$

Smoothed values: 45, 45, 45, 45, 45, 45, 45, 45, 45

Q10) Data Discretization: converts continuous data into a finite number of intervals.

Importance

1. Reduces noise
2. Simplifies analysis
2. Improves mining efficiency
3. Makes patterns easier to understand

equal-width	equal-frequency
1. Equal interval size	1. Equal number of records
2. Simple to implement	2. Better for skewed data
3. May create uneven bins	3. balanced distribution.

Impact on data analysis:

1. equal-width is suitable for uniformly distributed data
2. equal-frequency is better for skewed datasets.